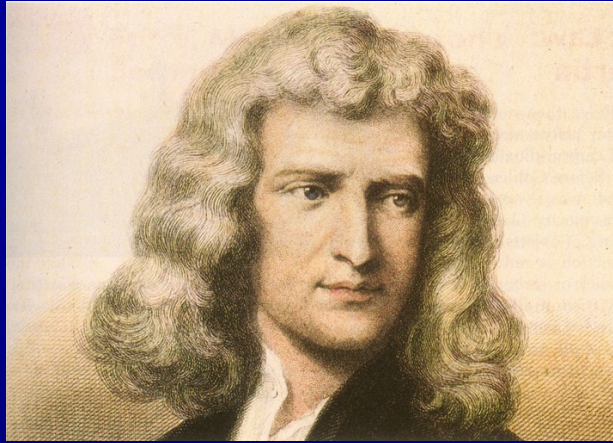


Chapter 3

Particle Dynamics

3.1 The law of **inertia**

Newton: 8 190 000



Newton's first law reads:

Every body persists in its state of rest or of uniform motion in a straight line unless it is compelled to change that state by forces impressed on it.

A free particle moves with constant velocity.

force-free

Force is negligible

Forces are cancelled

locally

$v = \text{const}$

including zero

Not true for some frame!

Inertial system? Stars?

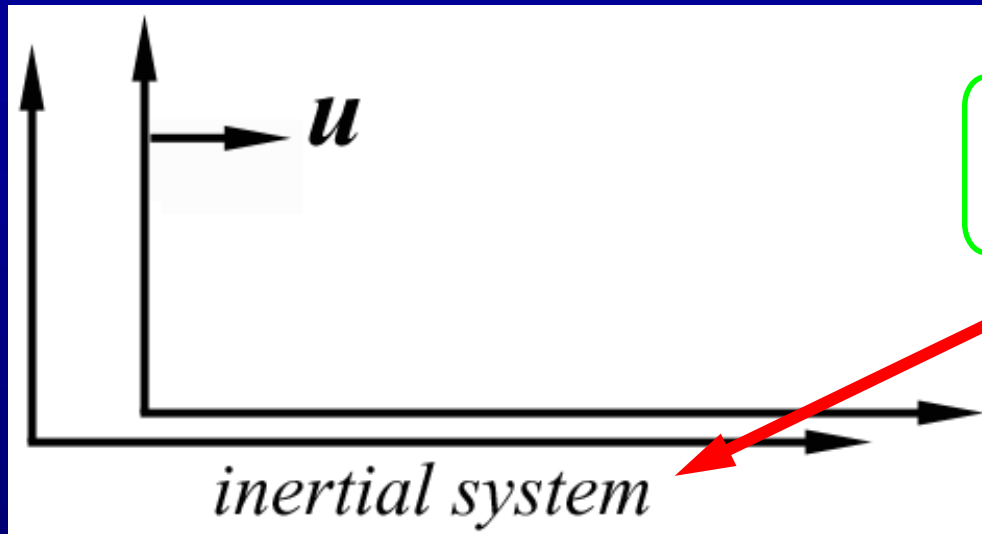
A reference frame in which **the first law is valid**
is called **inertial** or **Galilean system**.

Circular definition?

The 2007 Merriam-Webster dictionary defines a "hill" and a "mountain" this way:
hill - "1: a usually rounded natural elevation of land lower than a mountain"
mountain - "1a: a landmass that projects conspicuously above its surroundings
and is higher than a hill"

Circular reasoning (循环论证)

For a **free particle** in S



$$\dot{\mathbf{r}}' = \dot{\mathbf{r}} - \mathbf{u}$$

?

constant

constant

$$\dot{\mathbf{r}}' = \dot{\mathbf{r}} - \mathbf{u}$$

constant

constant

?

All inertial systems are equivalent.

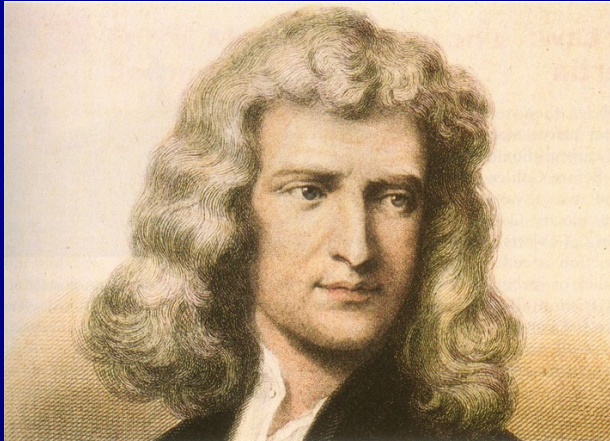
Earth  inertial system?

$$a = \frac{v^2}{R} = \omega^2 R = (2\pi/T)^2 R$$

earth's rotation	$3.4 \times 10^{-2} \text{m/s}^2$
earth's revolution	$5.9 \times 10^{-3} \text{m/s}^2$
sun's revolution	$3 \times 10^{-10} \text{m/s}^2$

Copernicus (1473—1543) (Heliocentric theory) asserts a better inertial system.

3.2 Newton's second and third laws



Newton's second law

$$\mathbf{F} = m\mathbf{a}$$

- The operational definition of force
- The definition of inertial mass

$$m_2 \mathbf{a}_2 = m_1 \mathbf{a}_1 \quad m_2 = m_1 |\mathbf{a}_1| / |\mathbf{a}_2|$$

- fundamental equation of dynamics

$$\mathbf{F} = m\mathbf{a}$$

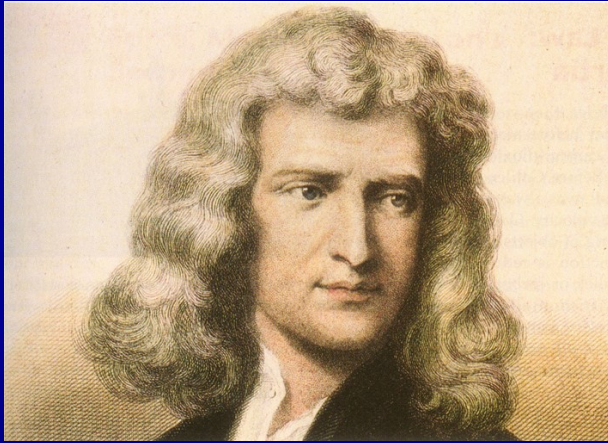
$$\mathbf{F} = \frac{d\mathbf{p}}{dt}$$

$$\mathbf{p} = m\mathbf{v} \quad \text{momentum}$$

$$m \frac{d}{dt} \mathbf{v} \neq \frac{d}{dt} \mathbf{p}$$

We may have $m = m(v)$

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$



Newton's third law

$$\mathbf{F}_{AB} = -\mathbf{F}_{BA}$$

- contact forces
- action-at-distance 超距作用

Modern viewpoint:

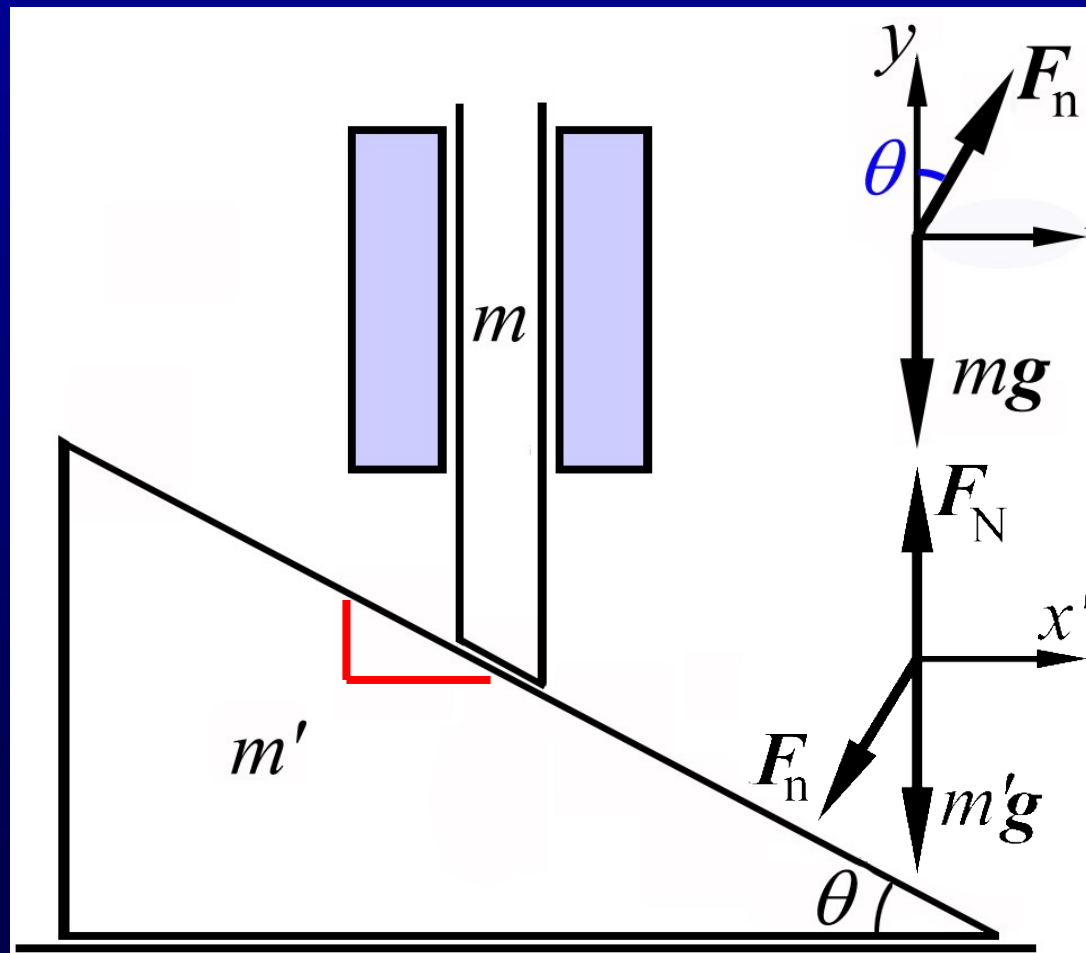
Interactions propagate through **fields** with **finite speed**.

problem-solving

- **body**
- **forces**
- **reference frame (origin, axes)**
- **free-body diagram**
- **equation of motion**
- **geometric relation**
- **approximation and solution**
- **discussion.**

Example 3.1

Find a' and F_n



$$F_n \cos \theta - mg = ma_y$$

$$F_N - F_n \cos \theta - m'g = 0$$

$$-F_n \sin \theta = m'a'_x$$

$$dy = dx' \tan \theta$$

$$a_y = a'_x \tan \theta$$

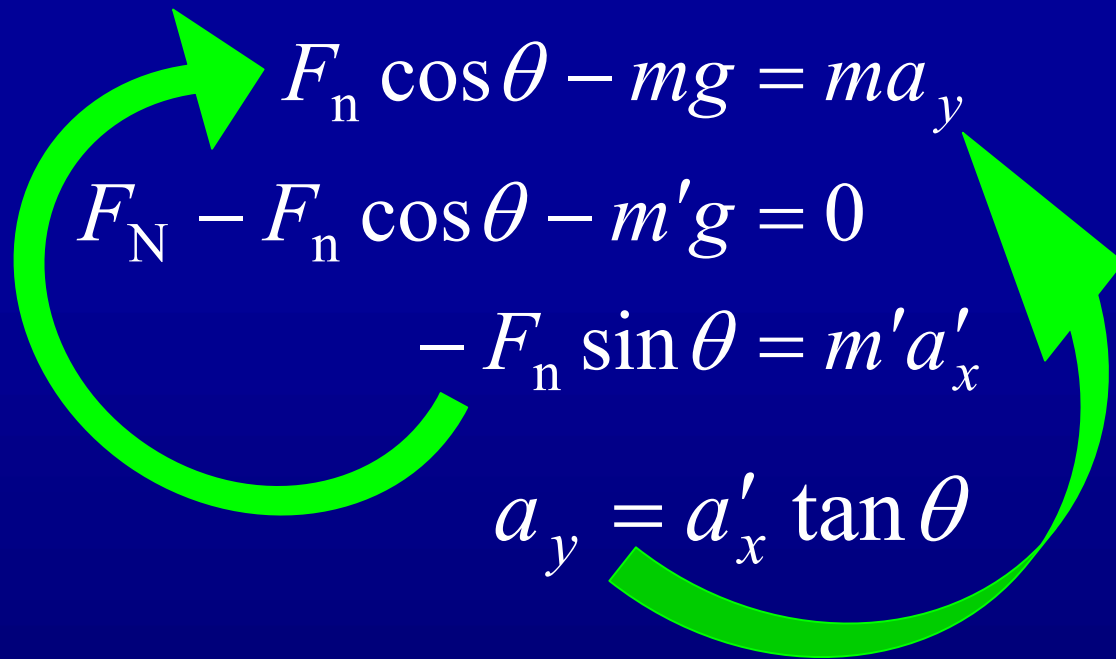


Diagram illustrating the relationship between accelerations a_y and a'_x using green curved arrows.

$$F_n \cos \theta - mg = ma_y$$

$$F_N - F_n \cos \theta - m'g = 0$$

$$-F_n \sin \theta = m'a'_x$$

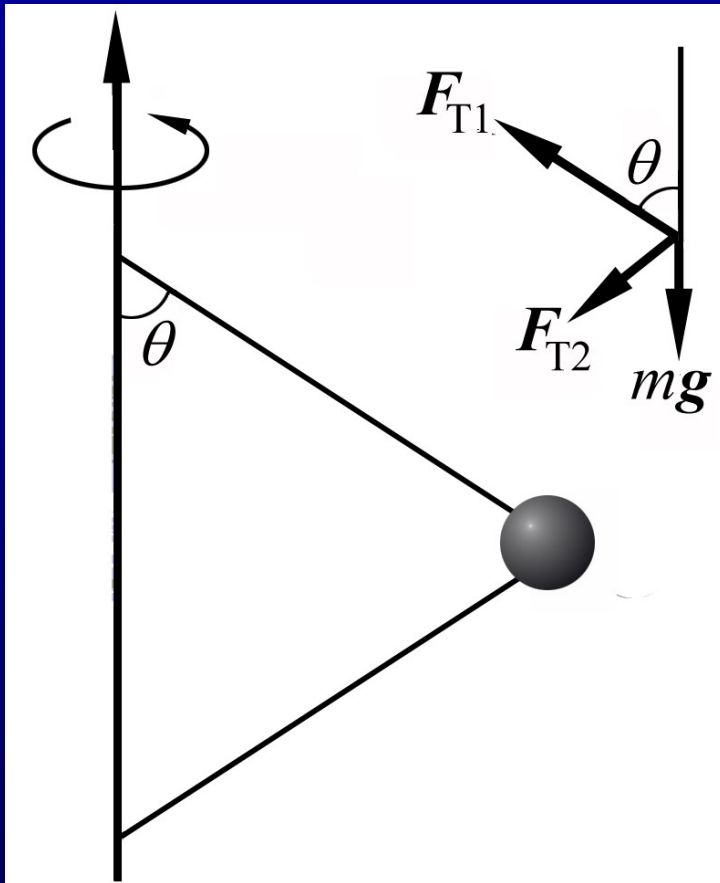
$$a_y = a'_x \tan \theta$$

$$-m'a'_x \cot \theta - mg = m \tan \theta a'_x$$

$$a'_x = -\frac{m}{m' \cot \theta + m \tan \theta} g$$

$$F_n = -\frac{m'}{\sin \theta} a'_x = \frac{mm' \cos \theta}{m' \cos^2 \theta + m \sin^2 \theta} g$$

Example 3.2



Known $m = 1.0\text{kg}$, $l = 1.0\text{m}$

$$F_{T1} = 25\text{N}, \quad \theta = 60^\circ$$

Find F_{T2} , v

Solution:

$$k : F_{T1} \cos \theta - F_{T2} \cos \theta - mg = 0$$

$$F_{T2} = F_{T1} - \frac{mg}{\cos \theta} = 5\text{N}$$

$$-e_\rho : F = (F_{T1} + F_{T2}) \sin \theta$$

$$= m \frac{v^2}{l \sin \theta}$$

$$v = \sqrt{\frac{(F_{T1} + F_{T2}) l \sin^2 \theta}{m}} = 4.7 \text{ m/s}$$

Simple pendulum

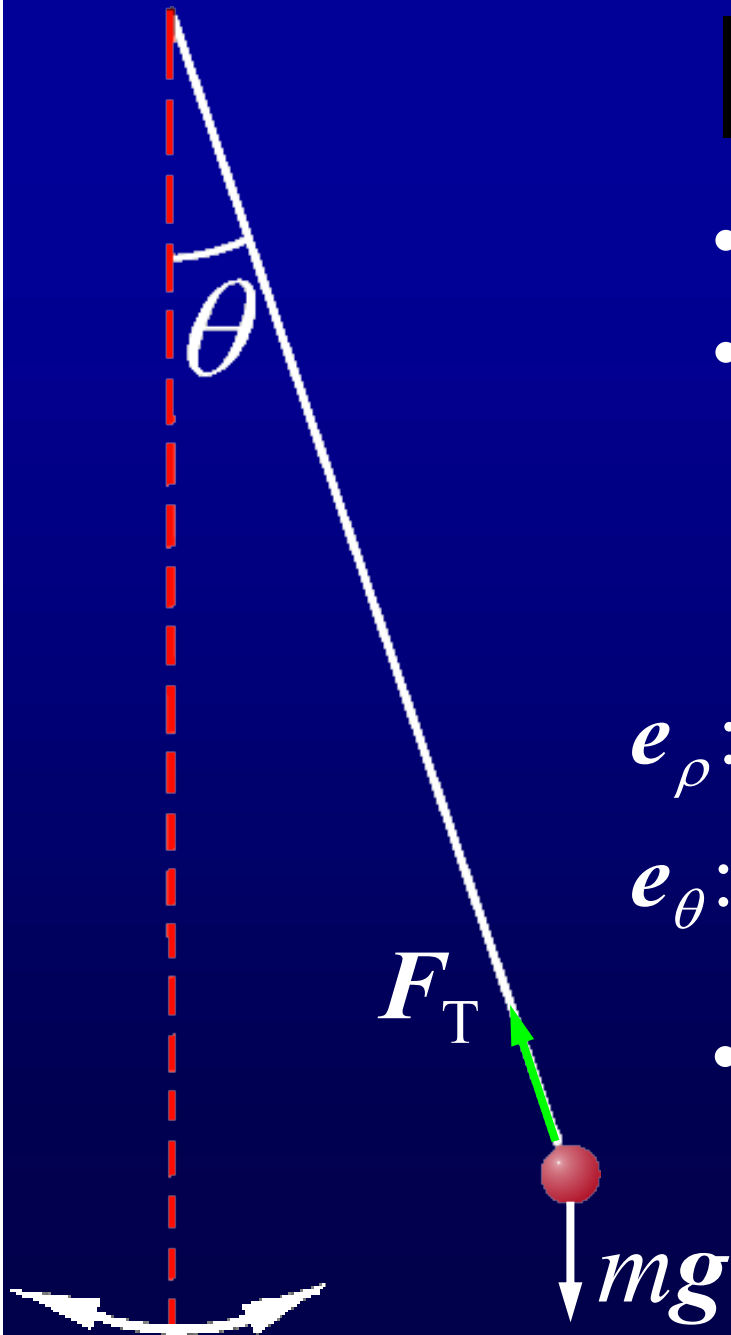
- a particle of mass
- a cord (or rod)
 - inextensible
 - of negligible mass

$$e_\rho: \quad mg \cos \theta - F_T = m(-L\dot{\theta}^2)$$

$$e_\theta: \quad -mg \sin \theta = mL\ddot{\theta}$$

- small angular displacement

$$\sin \theta \approx \theta$$



The equation in e_θ direction becomes

$$\ddot{\theta} + \left(\frac{g}{L} \right) \theta = 0$$

We may **write**

$$\theta = A \cos(\omega t + \varphi_0)$$

with A, ω, φ_0 to be fixed

- **initial conditions** $\theta(t=0) = \theta_0, \quad \dot{\theta}(t=0) = 0$

$$\theta = \theta_0 \cos(\omega t)$$

$$\omega^2 = \frac{g}{L}$$

$$-\omega A \sin(\varphi_0) = 0$$

$$\omega t = \sqrt{g/L} \ t = 2\pi \quad \text{defines period}$$

$$T = 2\pi \sqrt{\frac{L}{g}}$$

and frequency

$$f = \frac{1}{2\pi} \sqrt{\frac{g}{L}}$$

- **a particle of mass**
- **a cord (or rod)**
- **inextensible**
- **of negligible mass**
- **small angular displacement**
- **initial conditions**

For general case (angular displacement)

$$mL\ddot{\theta} = -mg \sin \theta$$

$$mL \frac{d\dot{\theta}}{dt} d\theta = -mg \sin \theta d\theta \quad d\theta = \dot{\theta} dt$$

$$\int_0^{\dot{\theta}} mL \dot{\theta} d\dot{\theta} = -mg \int_{\theta_0}^{\theta} \sin \theta d\theta$$

$$\dot{\theta}^2 = \frac{2g}{L} (\cos \theta - \cos \theta_0)$$

Comparing the equation

$$\ddot{\theta} + \left(\frac{g \sin \theta}{L \theta} \right) \theta = 0$$

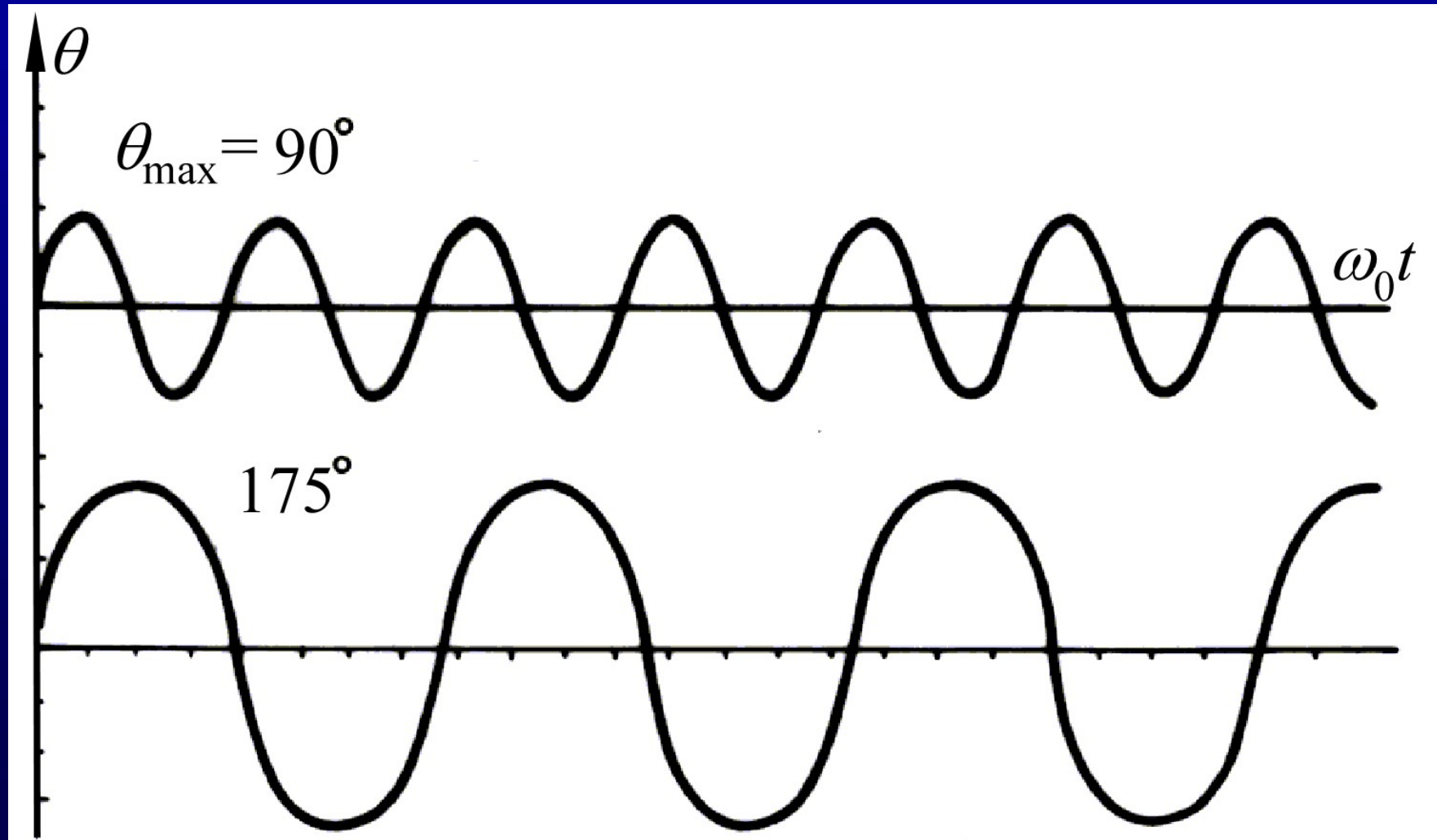
with **small angle** approximation equation

$$\ddot{\theta} + \left(\frac{g}{L} \right) \theta = 0$$

We may **feel**

$$“ \omega^2 \sim ” \frac{g \sin \theta}{L \theta} \leq \frac{g}{L} = \omega_0^2$$

Reliable conclusion comes from exact solution.



Assignment 3.1, 3.2

3.3 Forces

Gravitation

Chapter 4

- planet motion
- redshift of light in gravitational field
- accretion 吸积
- tide and tidal disruption
- ...

$$F_{21} = -G \frac{m_1 m_2}{r_{12}^2} \mathbf{r}_{12}$$

inverse
square law

Elastic or restoring force

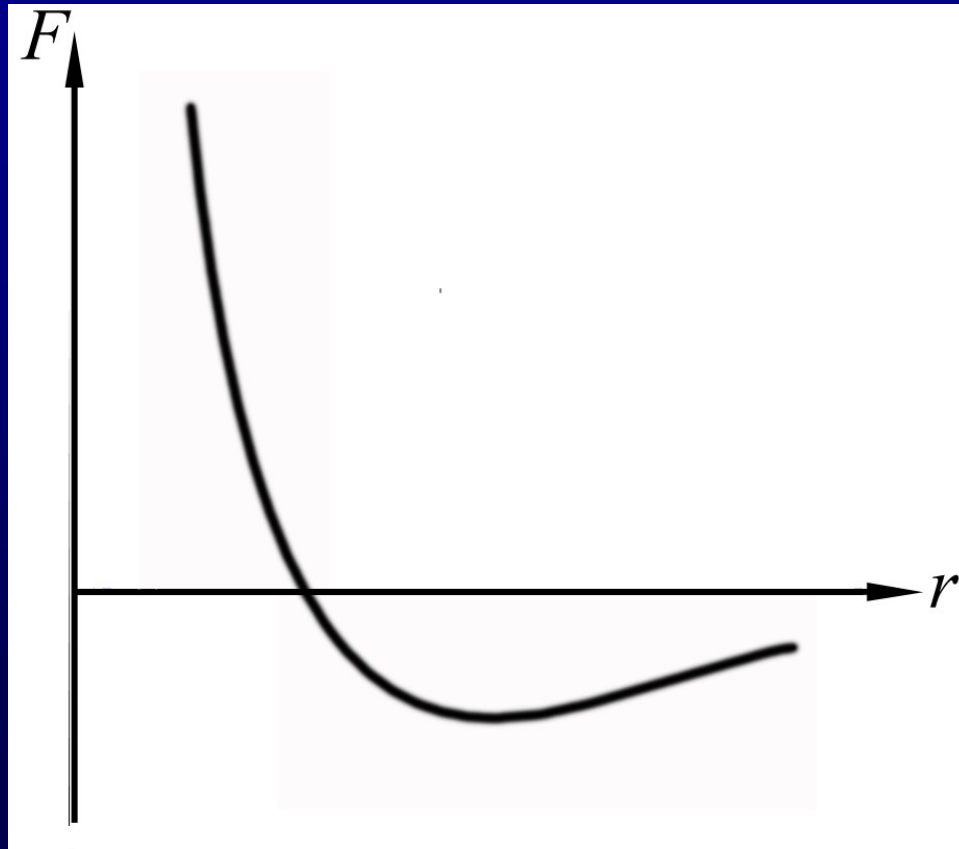
$$F = -kx \quad \text{Hooke's law}$$

Approximate, empirical

quark confinement? 10^{-15}m

Intermolecular force

$$F \sim 2\left(\frac{\sigma}{r}\right)^{13} - \left(\frac{\sigma}{r}\right)^7$$



complicated

**~ internal structure
of molecules**

Friction is the force between surfaces in contact that resists their relative tangential motion.

complicated → not fundamental

approximate, empirical laws

dissipative 耗散

industry and technology → Tribology

fundamental studies → dune model

(de Gennes) → self-organized criticality

Surface friction depends on many factors:

- **nature of materials**
- **surface finish** (表面光洁度)
- **surface (oxide) film**
- **extent of contamination**
- **temperature**
- ...

surface finishing tool

抛光工具

Numerically static friction ~ normal force

$$F_f \leq \mu_s F_N$$

coefficient
of static friction

Sliding friction ~ normal force

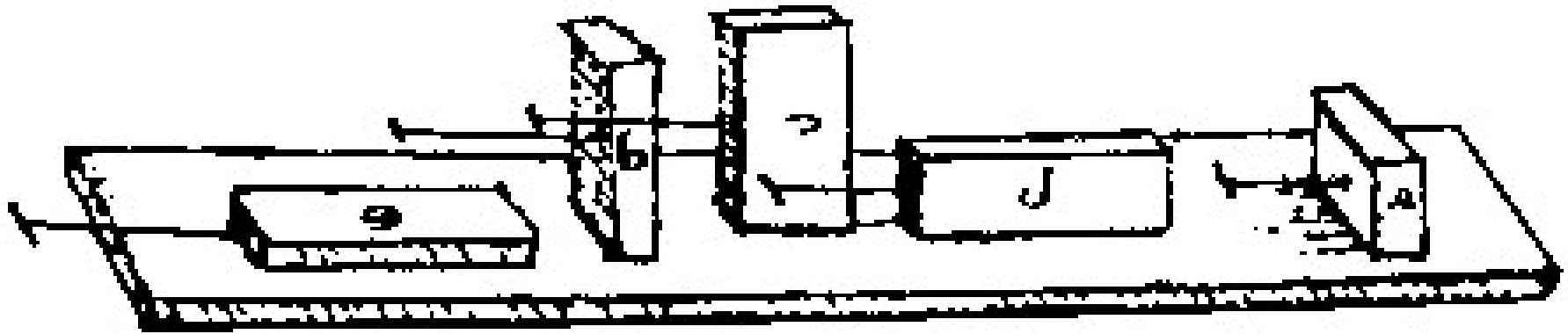
$$F_f = \mu_k F_N$$

coefficient
of kinetic friction

‘Amontons-Coulomb Law’

1699 1785

law of friction



One of Leonardo Da Vinci's sketches on friction. Among his conclusions from this experiment and others, we find the basic elements of the Amontons-Coulomb law.

3 centuries before Coulomb

Eric Gerde & M. Marder

Nature 413, 285–288 (20 September 2001)

"It is one of the dirty little secrets of physics that while we physicists can tell you a lot about quarks, quasars and other exotica, there is still no universally accepted explanation of the basic laws of friction."

cast iron ~ cast iron

$$\mu_s = 1.10 \quad \mu_k = 0.15$$

Teflon on Teflon

$$\mu_s \sim \mu_k \sim 0.04$$

friction piles

摩擦桩

end bearing piles

端承桩

Critical angle

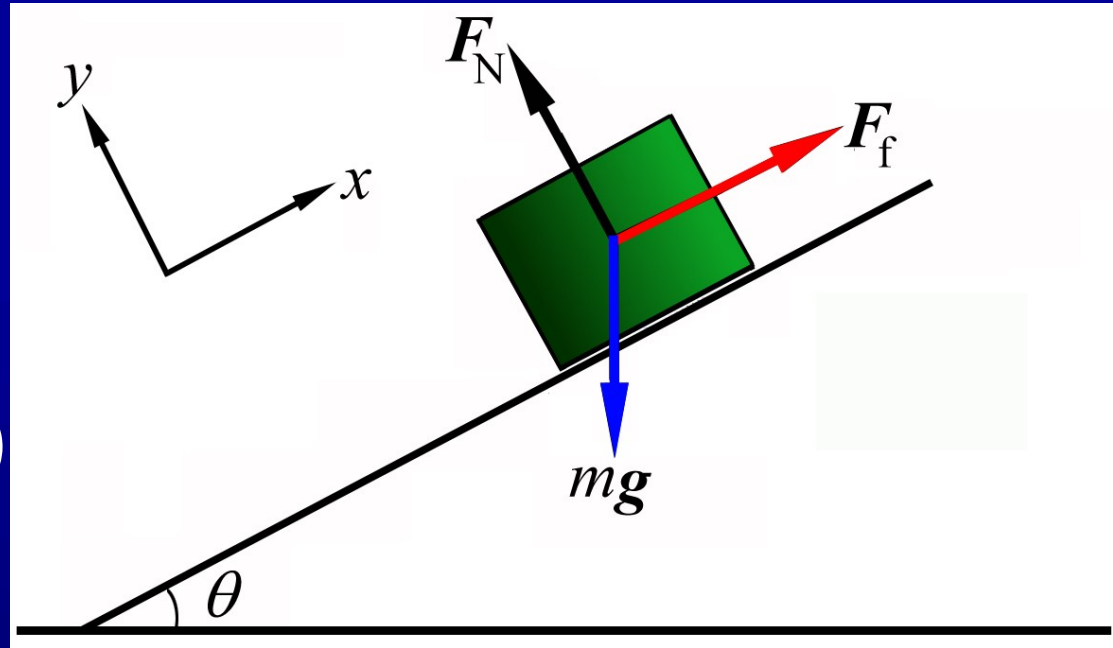
$$\mathbf{F_N} + \mathbf{F_f} + m\mathbf{g} = \mathbf{0}$$

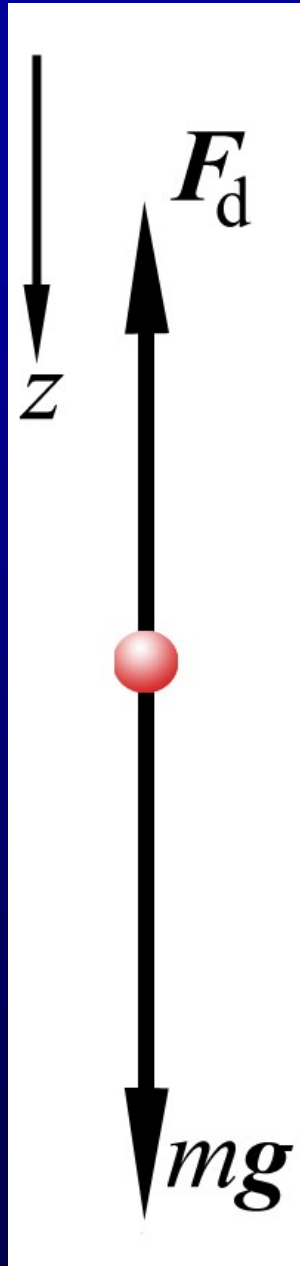
In x, y direction

$$F_f - mg \sin \theta = 0$$

$$F_N - mg \cos \theta = 0$$

$$\mu_s \geq \frac{F_f}{F_N} = \tan \theta \quad \text{i.e.} \quad \theta \leq \arctan \mu_s \equiv \theta_C$$





Example 3.3

Frictional drag and terminal speed

$$F_d = \frac{1}{2} C_d A \rho v^2$$

$$F_z = mg - \frac{1}{2} C_d A \rho v^2$$

$$F_z = 0 \quad \longrightarrow \quad v^2 = \frac{2mg}{C_d A \rho} \equiv v_T^2$$

$$F_z = mg \left(1 - \frac{v^2}{v_T^2} \right) \quad mg = \frac{1}{2} C_d A \rho v_T^2$$

$$\cancel{m} \frac{dv}{dt} = \cancel{mg} \left(1 - \frac{v^2}{v_T^2} \right)$$

$$\frac{dv}{dz} \frac{dz}{dt} = g \left(1 - \frac{v^2}{v_T^2} \right)$$

$$\frac{dv^2}{1 - v^2/v_T^2} = 2g dz$$

$$\boxed{-\ln \left(1 - \frac{v^2}{v_T^2} \right) \Big|_0^v = \frac{2g(z-0)}{v_T^2}}$$

$$v^2 = v_T^2 \left(1 - e^{-z/z_c} \right)$$

$$\boxed{\frac{2gz}{v_T^2} = \frac{z}{v_T^2/2g} \equiv \frac{z}{z_c}}$$

$$\frac{dv}{\left(1 - \frac{v^2}{v_T^2}\right)} = g dt$$

$$\left(\frac{1}{1 - \frac{v}{v_T}} + \frac{1}{1 + \frac{v}{v_T}} \right) d\left(\frac{v}{v_T} \right) = \frac{2g}{v_T} dt$$

$$\ln \frac{1 - \frac{v}{v_T}}{1 + \frac{v}{v_T}} = \frac{2g}{v_T} t$$

$$\frac{v}{v_T} = \frac{\exp\left(\frac{2g}{v_T}t\right) - 1}{\exp\left(\frac{2g}{v_T}t\right) + 1}$$

$$= \frac{\exp\left(\frac{g}{v_T}t\right) - \exp\left(-\frac{g}{v_T}t\right)}{\exp\left(\frac{g}{v_T}t\right) + \exp\left(-\frac{g}{v_T}t\right)}$$

$$v = v_T \tanh \frac{g}{v_T} t \quad \text{hyperbolic tangent}$$

Viscosity force

$$F_{\eta} = 6\pi\eta rv$$

coefficient of viscosity
粘滞系数

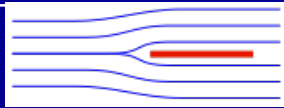
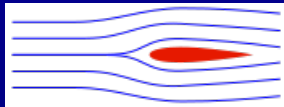
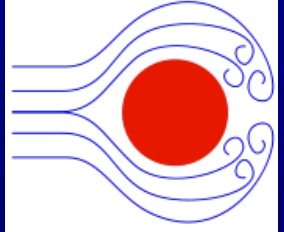
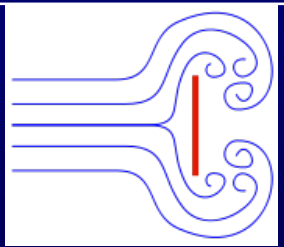
$$1\text{N}/[\text{m}(\text{m}\cdot\text{s}^{-1})] = 1\text{Pa}\cdot\text{s}$$

Reynolds number

$$Re = \frac{\rho vd}{\eta} \sim 0 - 10$$

Drag (fluid resistance)

[https://en.wikipedia.org/wiki/Drag_\(physics\)](https://en.wikipedia.org/wiki/Drag_(physics))

Shape of flow	Form drag	Skin friction	Drag force
	0%	100%	laminar drag $F \propto v$
	~10%	~90%	
	~90%	~10%	
	100%	0%	turbulent drag $F \propto v^2$

For dry friction, the resistance force is nearly independent of the velocity. (Amontons-Coulomb law)

~ (tilde)--- approximately, around, about, of the same order of magnitude (数量级)

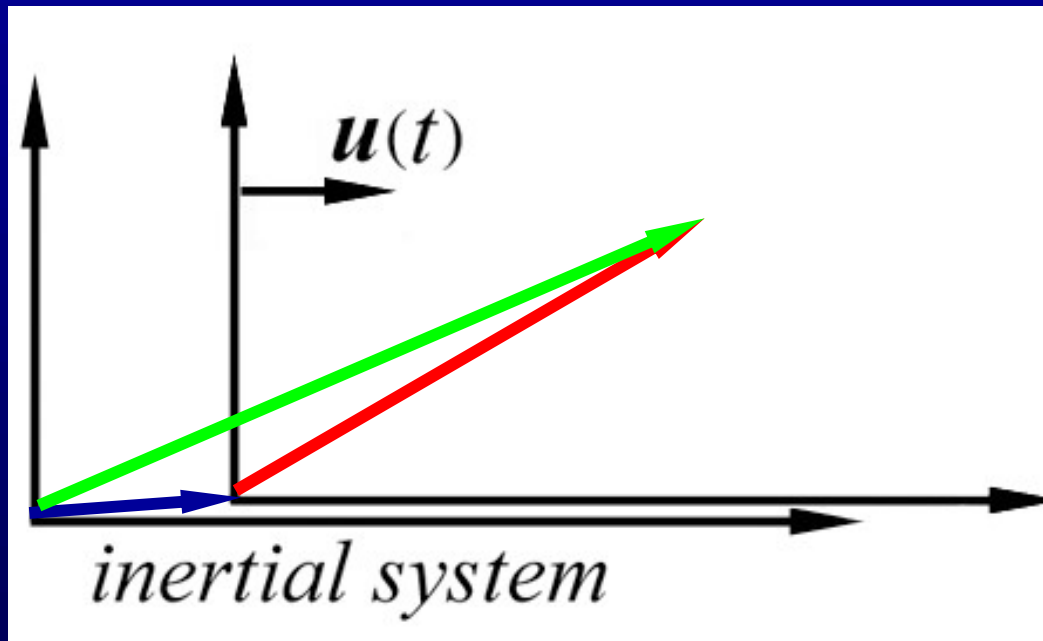
Nuclear force (between nucleons)

$$F \sim \frac{1}{r^2} \exp(-r/r_0), \quad r_0 \sim 10^{-14} \text{ m}$$

Nucleons have structure.

3.4 Noninertial frame and inertial force

For a free particle in S,



$$\mathbf{r}'(t) = \mathbf{r} - \overrightarrow{OO'}$$

$$\mathbf{v}'(t) = \mathbf{v} - \mathbf{u}(t)$$

In S' 1st law $\times$

S' is called as **noninertial frame**.

Secondly if $F \neq 0$,

$$\ddot{\mathbf{r}} = \mathbf{a}' + \ddot{\mathbf{u}}$$

$$\mathbf{F} = m\mathbf{a}' + m\ddot{\mathbf{u}}$$

$$\mathbf{F}_{\text{in}} \equiv -m\ddot{\mathbf{u}}$$

In S' 2nd law $\times$

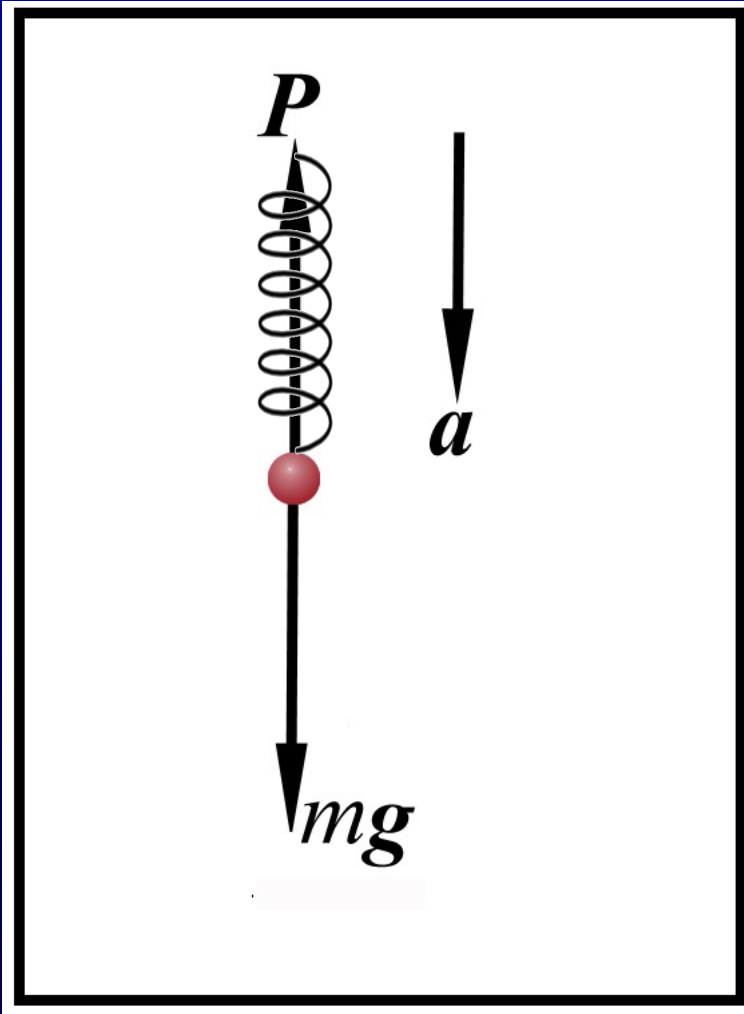
$$(\mathbf{F} + \mathbf{F}_{\text{in}}) = m\mathbf{a}'$$

In S' 2nd law $\checkmark$

$\mathbf{F}_{\text{in}}$ is called as **inertial force**

视重

Example 3.4 Apparent weight



Equation of motion in S

$$m_g g + P = m_i a$$

Apparent weight:

$$P = \left| -(m_g g - m_i a) \right|$$

In S'

$$m_g g + P + (-m_i a) = 0$$

acceleration of the elevator

Weightlessness In free-falling elevator

$$P = \left| - (m_g g - m_i a) \right| = 0$$

when $m_g = m_i$ $g = a$

Gravitation can be canceled

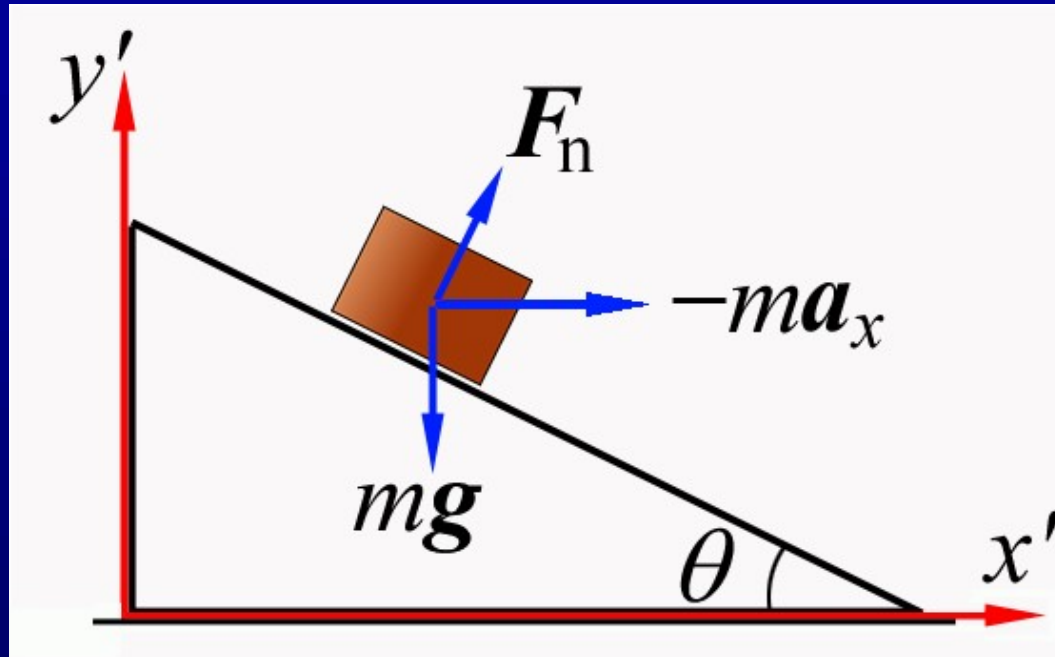
We can not distinguish

- gravitation and inertial force
- gravitational field and accelerated frame

principle of equivalence 

General theory of relativity

Example 3.5



In wedge frame ($\ddot{x}$)

$$F_n \sin \theta - m\ddot{x} = m\ddot{x}'$$

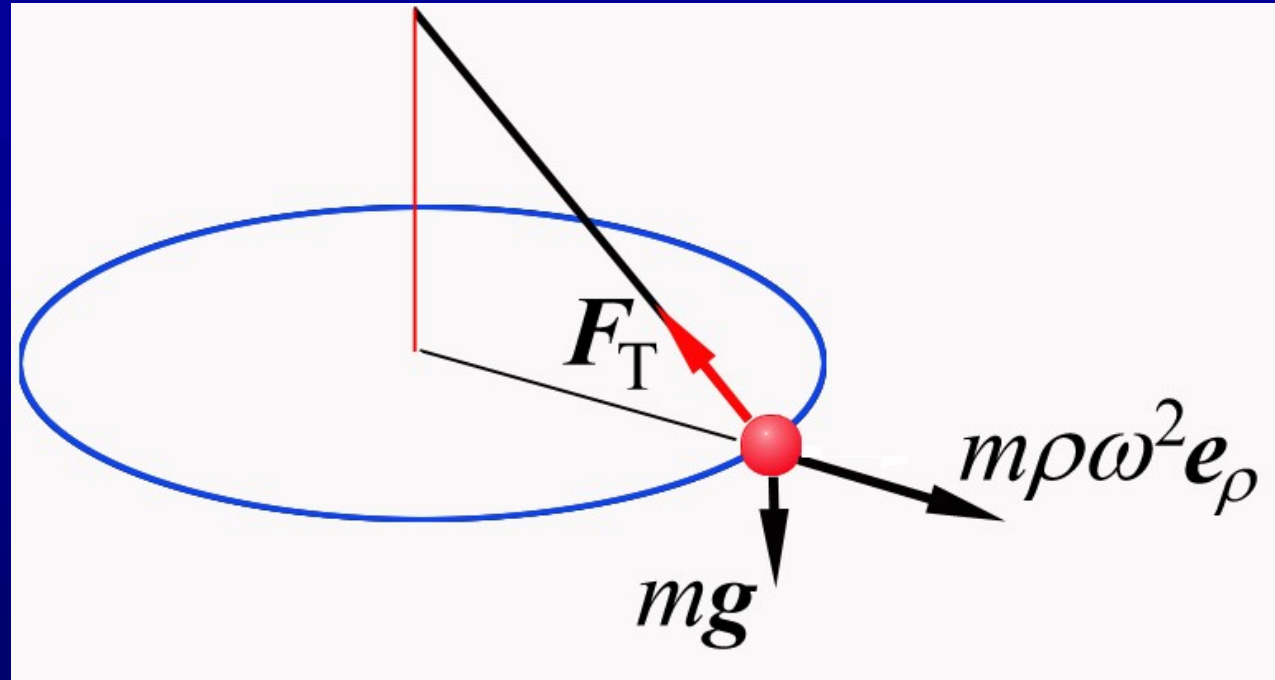
$$F_n \cos \theta - mg = m\ddot{y}'$$

$$-dy' = dx' \tan \theta$$

Only in S'

Complete solution needs
equation of motion for wedge

Conic pendulum



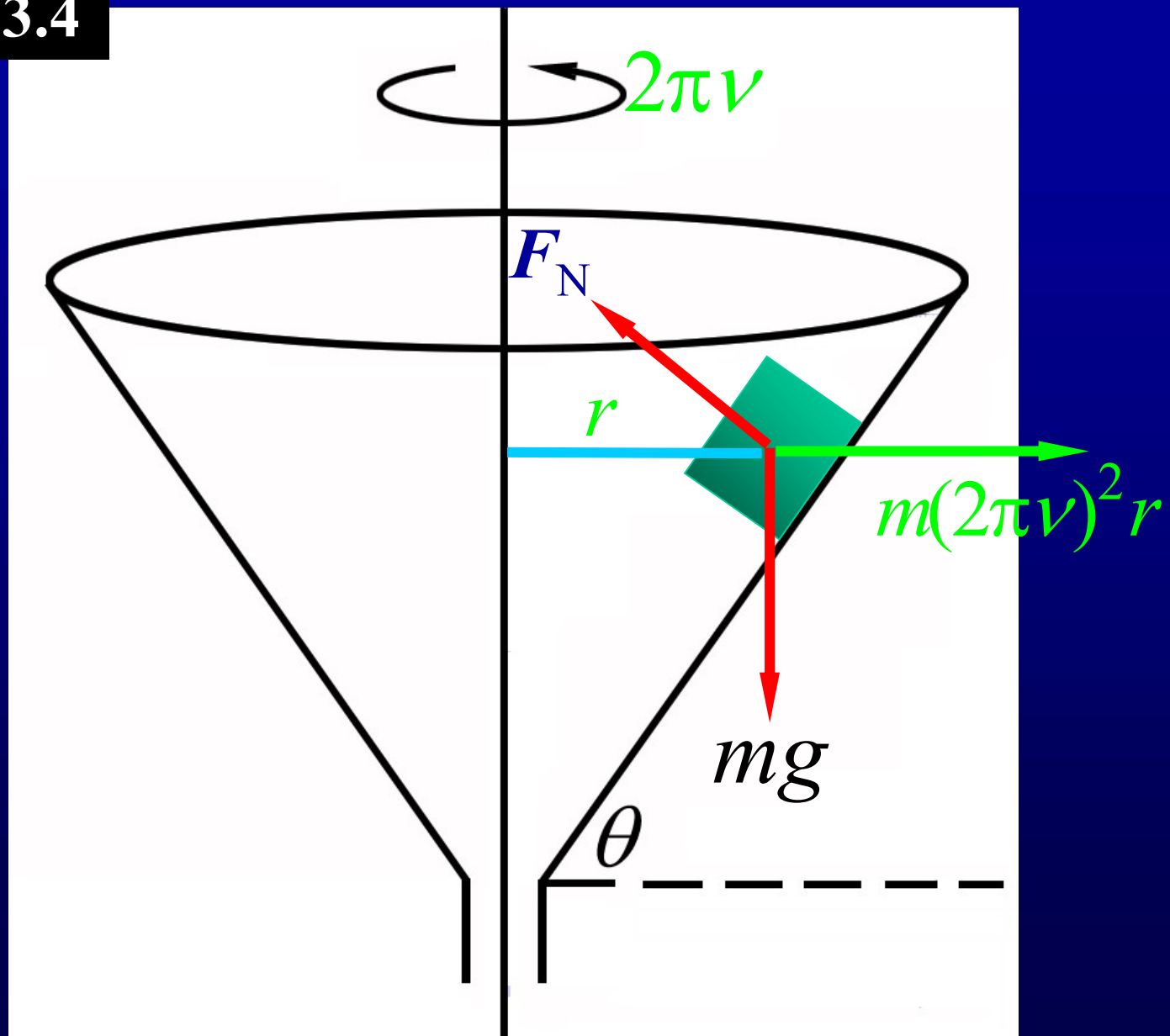
S

$$F_T + mg = -m\rho\dot{\phi}^2 e_\rho$$

$$F_T + mg + m\rho\dot{\phi}^2 e_\rho = \mathbf{0}$$

S'

Problem 3.4



Coriolis force

$$m\mathbf{a}_{\text{rot}} = m\mathbf{a}_{\text{in}} - \underline{2m\boldsymbol{\omega} \times \dot{\mathbf{r}}_{\text{rot}}} - m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$$

inertial centrifugal force

kinematic effect

Newton's	second law	valid
	third law	not valid

Assignment: 3.5, 3.4, 3.7/*3.9

" What Descartes did was a good step. You have added much several ways, and especially in taking the colours of thin plates into philosophical consideration. If I have seen a little further it is by standing on the shoulders of Giants."

Isaac Newton's letter to his rival Robert Hooke that was dated February 5, 1676 .

sarcastic remark ?

Dry Friction due to Adsorbed Molecules

C. Daly, J. Zhang, and J. B. Sokoloff

Phys. Rev. Lett. 90, 246101 (2003)

hyperbolic functions

Hyperbolic sine

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

Hyperbolic cosine

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

Hyperbolic tangent

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

3.5 Momentum and Angular Momentum

$$\mathbf{p} = m\mathbf{v}$$

$$\mathbf{F} = \frac{d\mathbf{p}}{dt}$$

$$\mathbf{F} = \mathbf{0}, \quad \mathbf{p} = \mathbf{C}$$

$$F_i = 0, \quad p_i = C_i$$

Conservation of momentum


$$\mathbf{p} = \frac{m_0 \mathbf{v}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

discoveries $n \rightarrow p + e + \bar{\nu}$

The conservation of momentum and energy requires a third particle.

In 1930 Pauli proposed the existence of an “undetectable, light, and neutral” particle.

In 1934 E. Fermi named it as “**neutrino**”

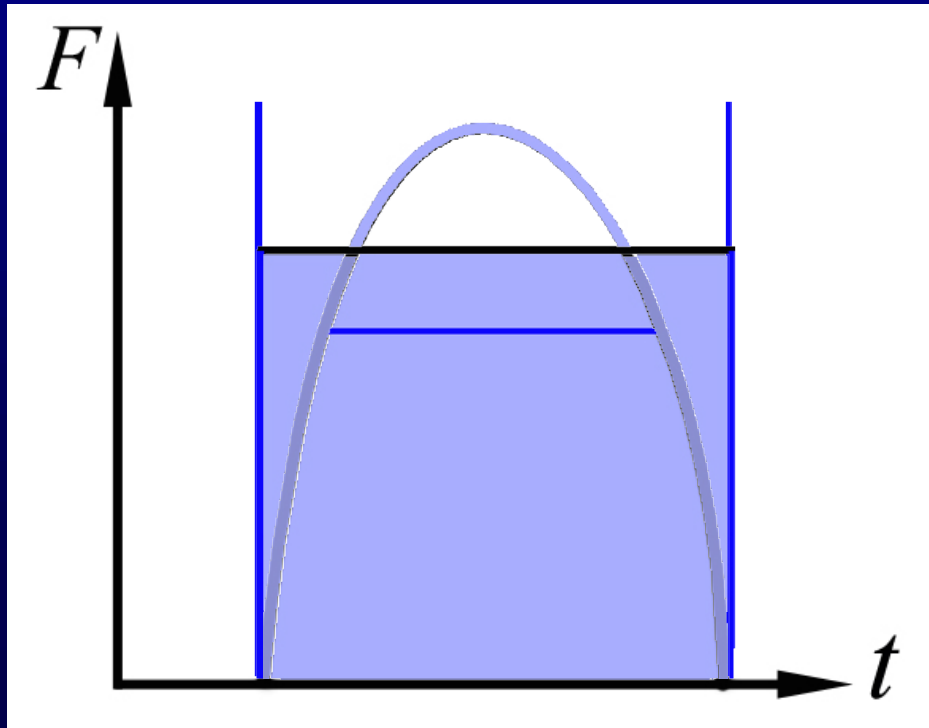
In 1942, Kan-Chang Wang(王淦昌) first proposed the use of K-capture to experimentally detect neutrinos.

In 1956 C. L. Cowan Jr. and F. Reines et al. detected neutrinos

$$\mathbf{p}_f - \mathbf{p}_i = \int_{\mathbf{p}_i}^{\mathbf{p}_f} d\mathbf{p} = \int_{t_i}^{t_f} \mathbf{F} dt \equiv \mathbf{I} = \overline{\mathbf{F}} \times (t_f - t_i)$$

Impulse: time integral of the force

冲量



$$\overline{\mathbf{F}} \equiv \frac{1}{t_f - t_i} \int_{t_i}^{t_f} \mathbf{F} dt$$

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}$$

angular momentum **with respect to O**

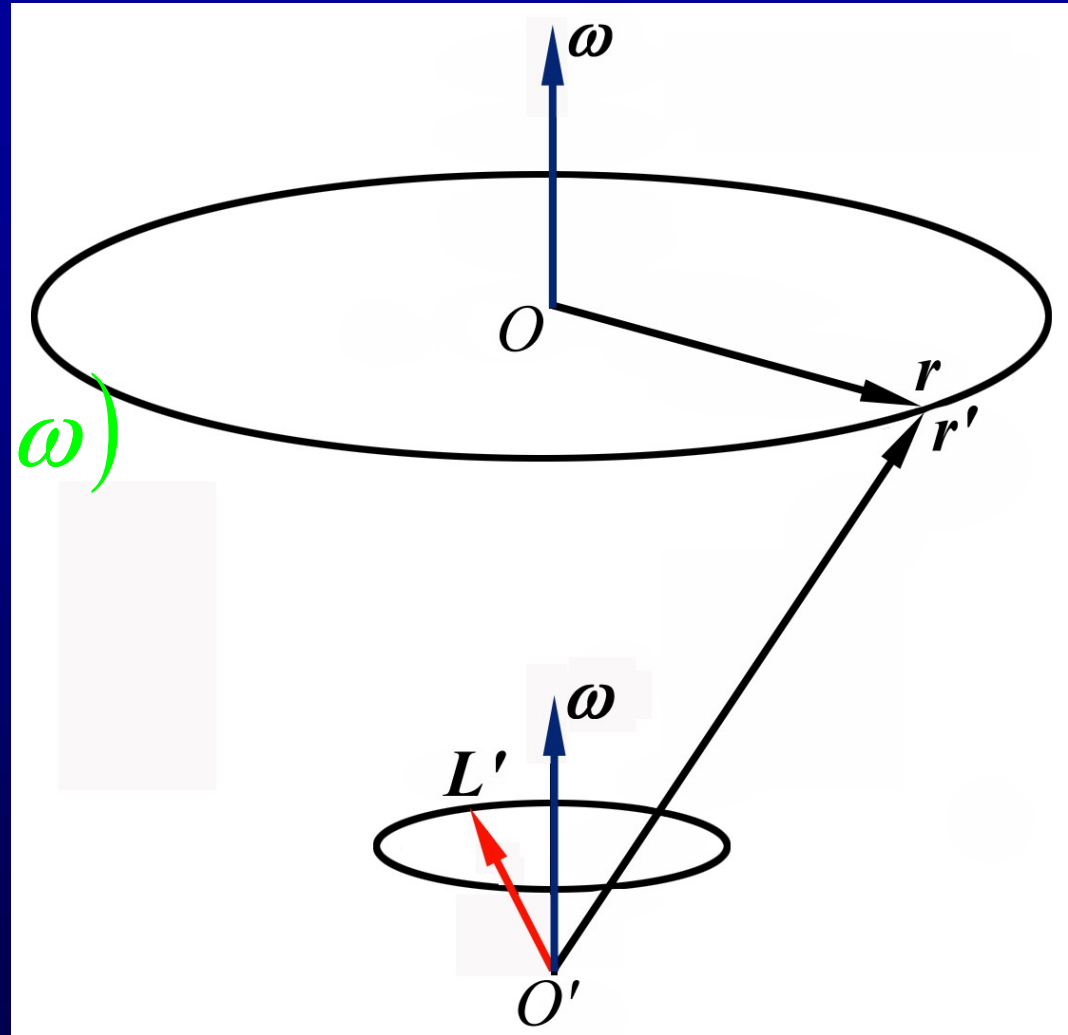
In a circular motion

$$\begin{aligned} \mathbf{L} &= m\mathbf{r} \times (\boldsymbol{\omega} \times \mathbf{r}) \\ &= mr^2\boldsymbol{\omega} - m\mathbf{r}(\mathbf{r} \cdot \boldsymbol{\omega}) \end{aligned}$$

With respect to O

$$\mathbf{L} = \boxed{mr^2}\boldsymbol{\omega}$$

$$\sim \mathbf{p} = m\mathbf{v}$$



Bohr model of hydrogen atom

$$m_e = 9.11 \times 10^{-31} \text{ kg} \quad \langle r \rangle = 0.529 \times 10^{-10} \text{ m}$$

$$\omega = 4.13 \times 10^{16} \text{ s}^{-1}$$

$$L = 1.05 \times 10^{-34} \text{ m}^2 \text{ kg/s} = \hbar$$

Earth-moon system

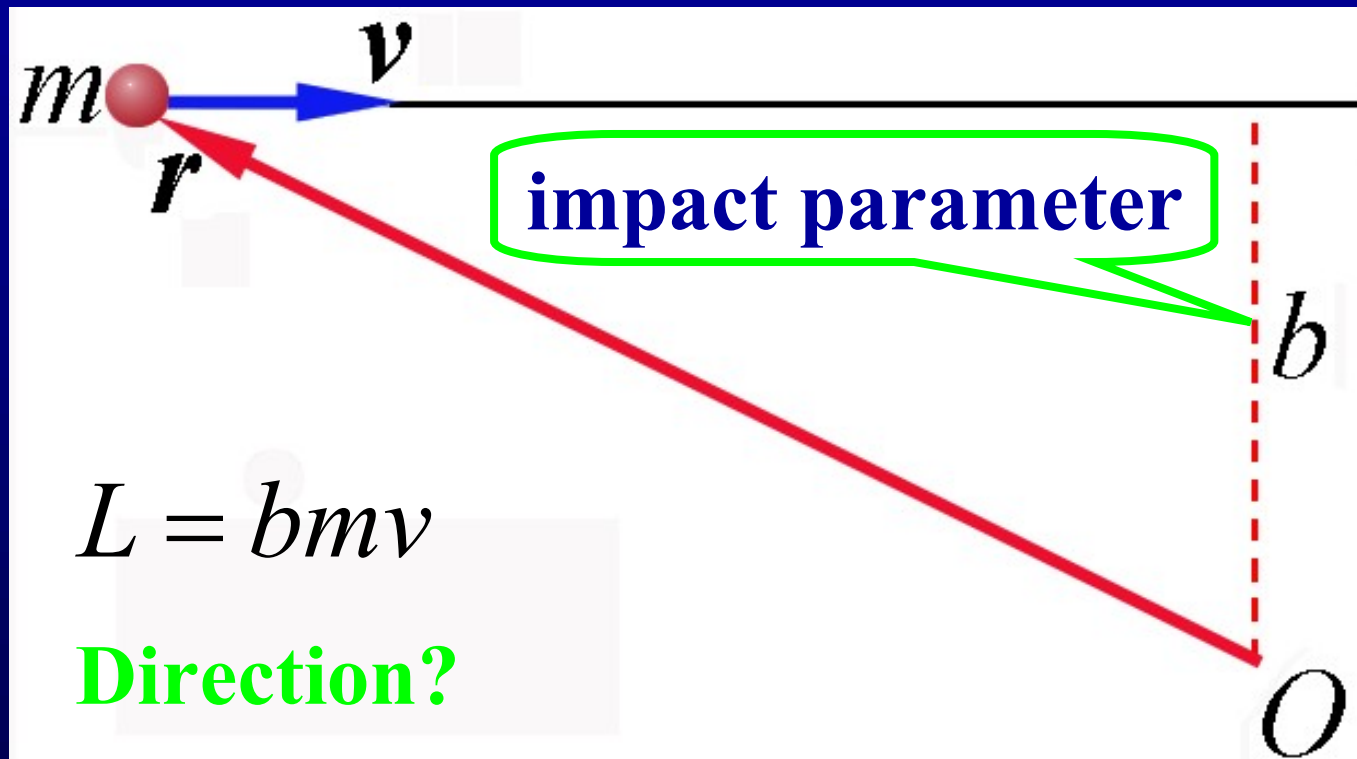
$$m = 7.36 \times 10^{22} \text{ kg} \quad \bar{r} = 3.84 \times 10^8 \text{ m} \sim 60R_{\oplus}$$

$$T = 27.3 \times 8.616 \times 10^4 \text{ s}$$

$$L = 2.90 \times 10^{34} \text{ m}^2 \text{ kg/s}$$

Angular momentum exists for finite r, p .

Example 3.6



Change rate of angular momentum

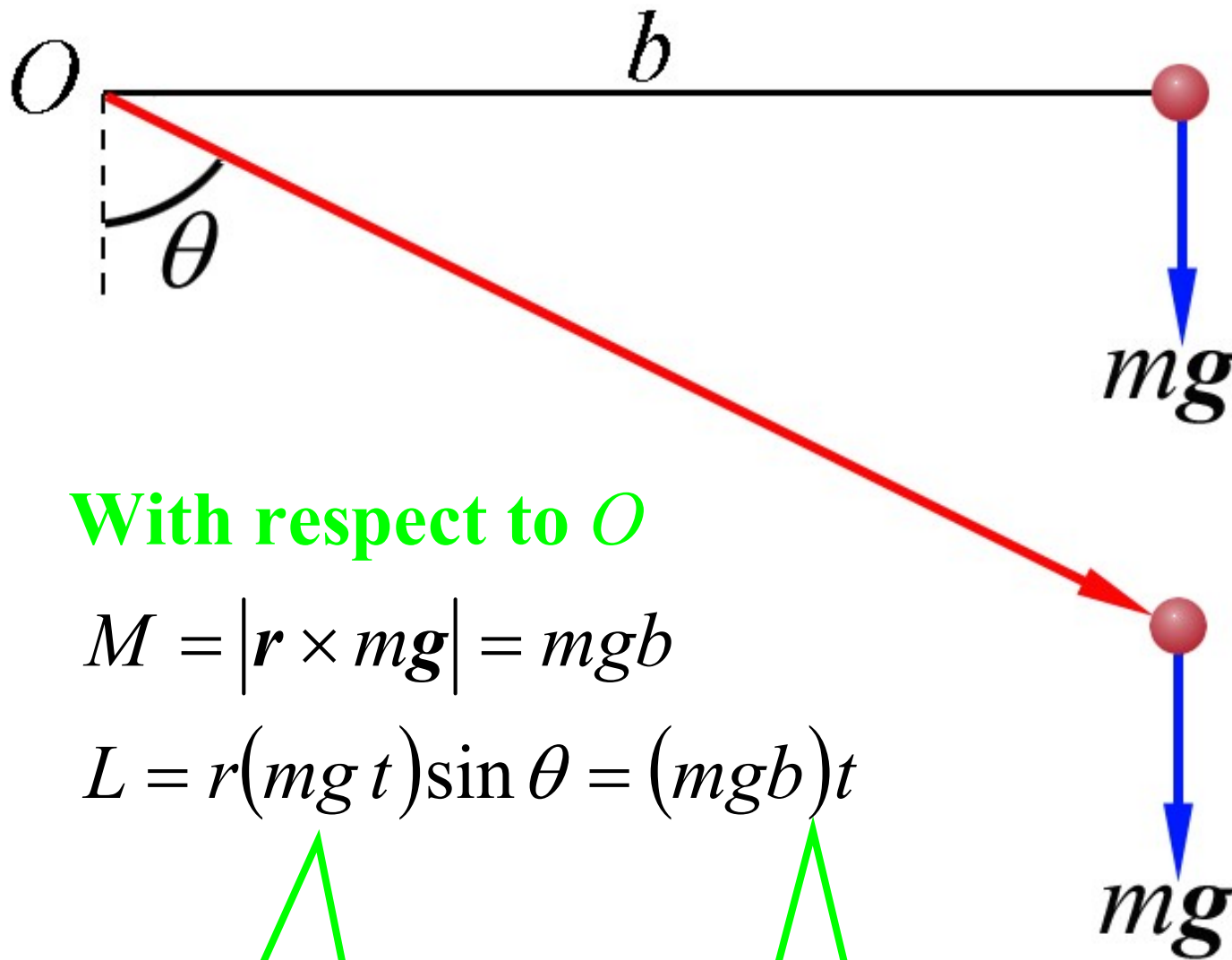
$$\frac{d\mathbf{L}}{dt} = \frac{d\mathbf{r}}{dt} \times \mathbf{p} + \mathbf{r} \times \frac{d\mathbf{p}}{dt}$$

$$= \mathbf{r} \times \mathbf{F}$$

$$= \mathbf{M}$$

torque with respect to O

力矩



$$\frac{dL}{dt} = M$$

With respect to O

$$M = |\mathbf{r} \times m\mathbf{g}| = mgb$$

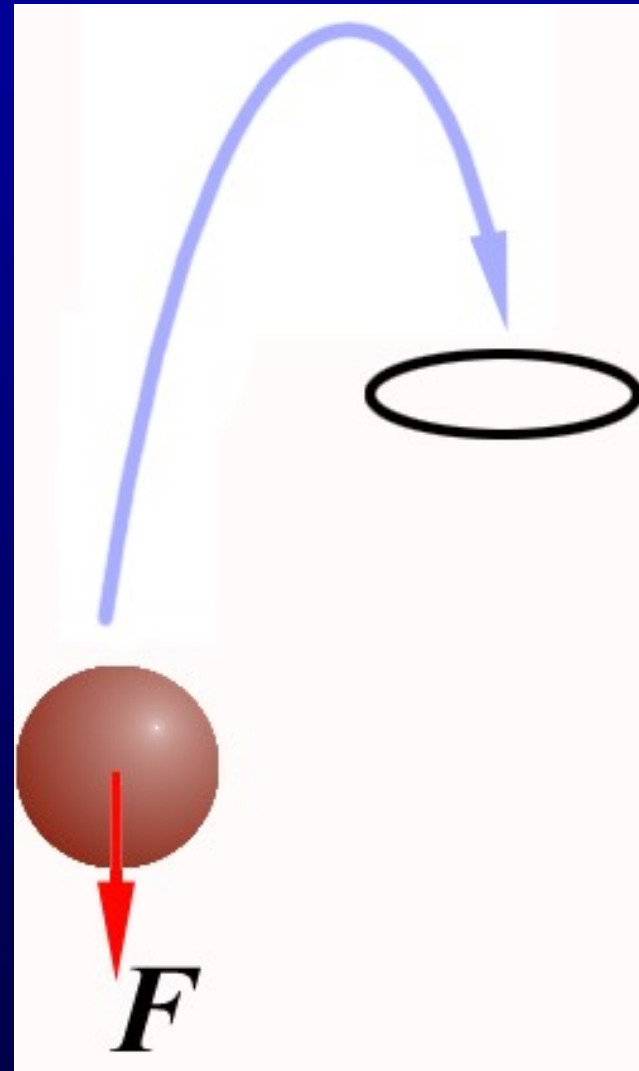
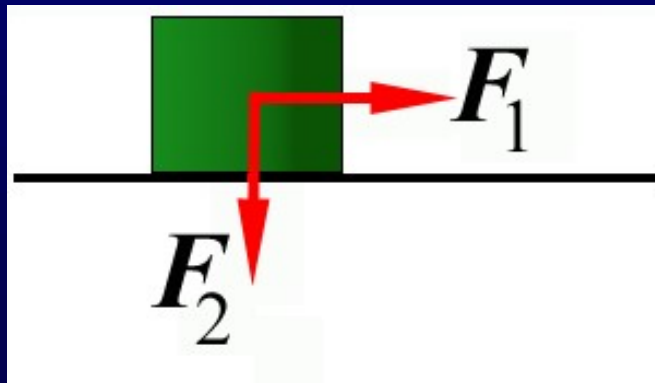
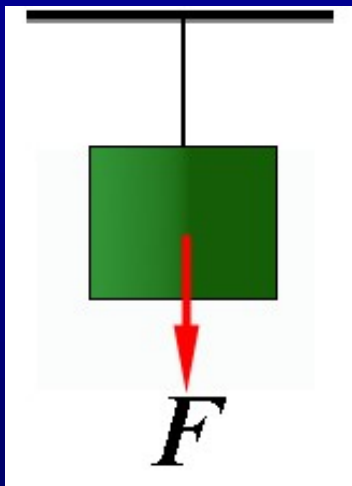
$$L = r(mg t) \sin \theta = (mgb)t$$

$$rp \sin \theta$$

$$\int M dt$$

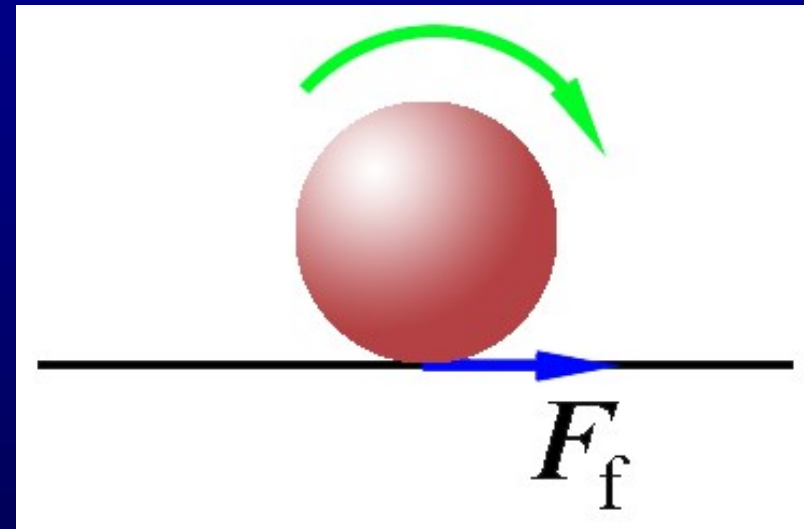
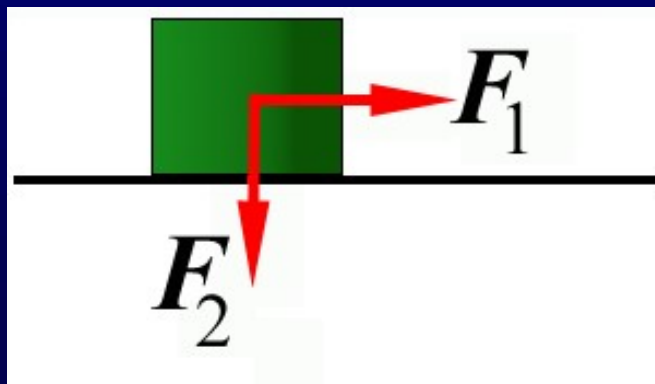
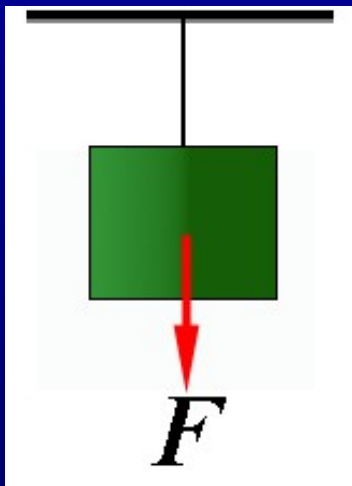
3.6 Mechanical Work and Energy

$$dW = \mathbf{F} \cdot d\mathbf{r}$$



3.6 Mechanical Work and Energy

$$dW = \mathbf{F} \cdot d\mathbf{r}$$



spatial integral of the force

work-energy relation

$$\begin{aligned} W &= \int \mathbf{F} \cdot d\mathbf{r} = \int m \frac{d\mathbf{v}}{dt} \cdot \frac{d\mathbf{r}}{dt} dt \\ &= \int m \frac{d\mathbf{v}}{dt} dt \cdot \frac{d\mathbf{r}}{dt} \\ &= \int m d\mathbf{v} \cdot \frac{d\mathbf{r}}{dt} \\ &= \int m \mathbf{v} \cdot d\mathbf{v} \end{aligned}$$

work-energy relation

$$\begin{aligned} W &= \int \mathbf{F} \cdot d\mathbf{r} = \int m\mathbf{v} \cdot d\mathbf{v} = \int_{v_i}^{v_f} m v dv = \frac{1}{2} m v^2 \Big|_{v_i}^{v_f} \\ &= \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 \quad \mathbf{v} \cdot \mathbf{v} = v^2, \quad 2\mathbf{v} \cdot d\mathbf{v} = 2v dv \\ &\equiv T_f - T_i \quad T = \frac{1}{2} m v^2 + C \\ &= \Delta T \end{aligned}$$

kinetic energy

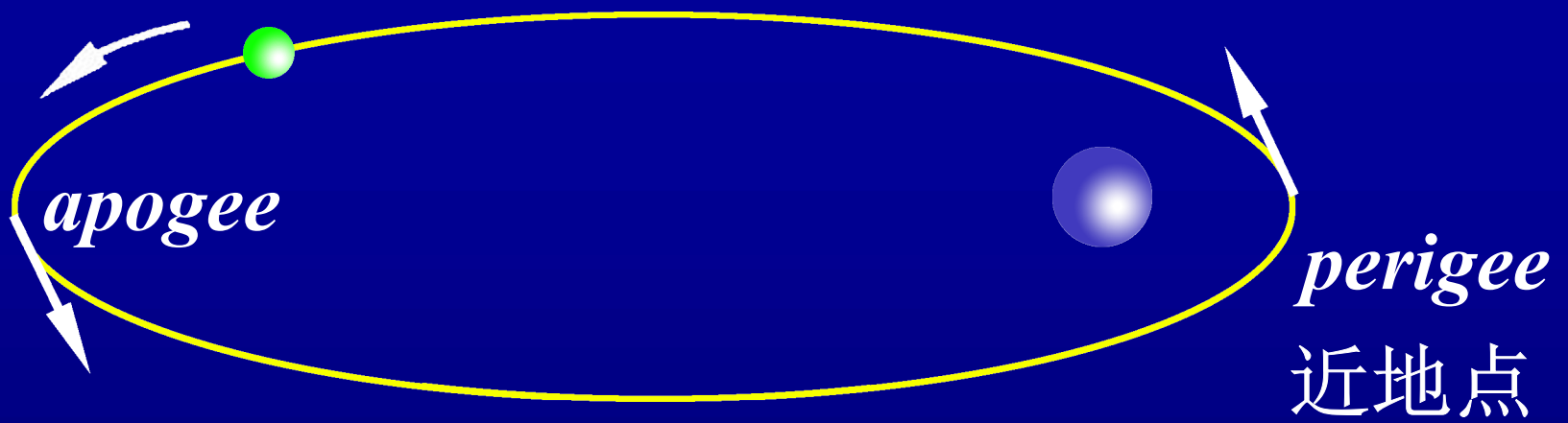
increment of kinetic energy

Consider a system S'

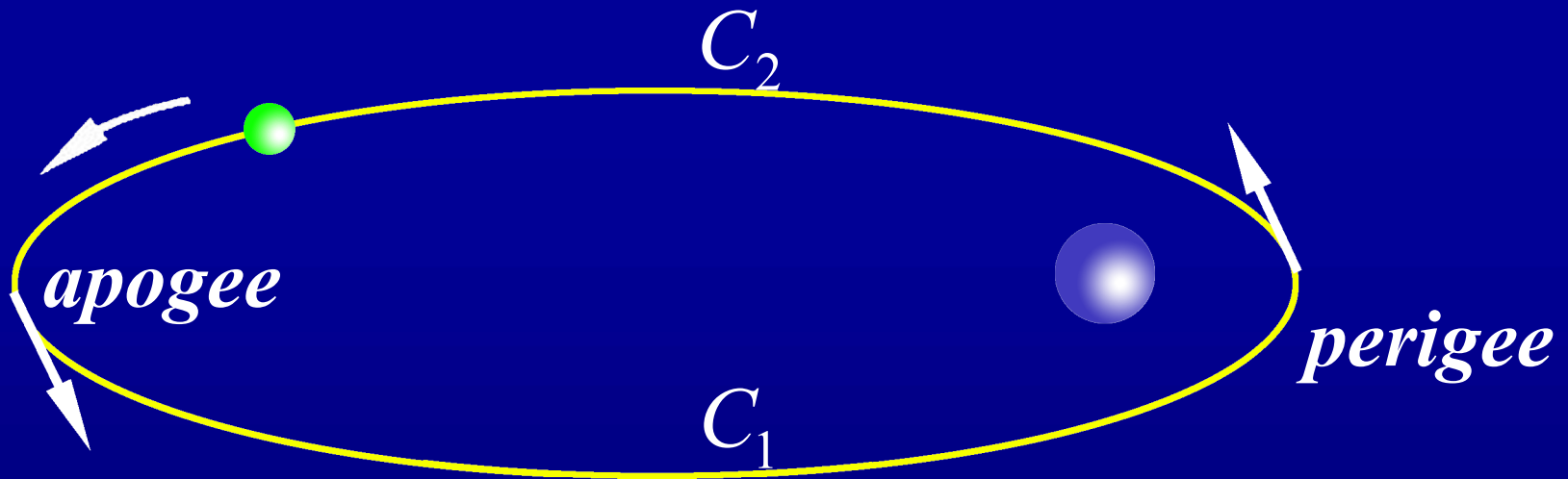
$$\begin{aligned}\frac{1}{2}mv^2 &= \frac{1}{2}m(\mathbf{v}' + \mathbf{u}) \cdot (\mathbf{v}' + \mathbf{u}) \\ &= \frac{1}{2}mv'^2 + \frac{1}{2}m(2\mathbf{v}' + \mathbf{u}) \cdot \mathbf{u}\end{aligned}$$

Kinetic energy depends on **reference frame**!

Does **work** depend on **reference frame**?



$$T_i = T_f \longrightarrow W_{apa} = W_{pap} = 0$$



$$\oint F \cdot dr = 0$$

$$\int_a^p F \cdot dr + \int_p^a F \cdot dr = 0$$

conservative

Along C_1

Along C_2

**Around the sun:
perihelion
aphelion**

$$\int_{r_i}^r \mathbf{F} \cdot d\mathbf{r} \equiv -U(\mathbf{r}) \Big|_{r_i}^r = U(\mathbf{r}_i) - U(\mathbf{r}) = -\Delta U$$

$$U = mgh + C$$

potential energy **function**

$$\int_A^x (-kx) dx = -\left(\frac{1}{2} kx^2 - \frac{1}{2} kA^2 \right)$$

$$U = \frac{1}{2} kx^2 + C$$

additive constant

$$\int_{r_i}^r \mathbf{F} \cdot d\mathbf{r} = U(\mathbf{r}_i) - U(\mathbf{r})$$

$$\mathbf{1D} \quad \int_{x_i}^x F \cdot dx = U(x_i) - U(x) \quad F_x = -\frac{dU}{dx}$$

$$\mathbf{3D} \quad \longrightarrow \mathbf{F} = -\nabla U$$

where

$$\mathbf{gradient} \quad \nabla U \equiv \mathbf{i} \frac{\partial U}{\partial x} + \mathbf{j} \frac{\partial U}{\partial y} + \mathbf{k} \frac{\partial U}{\partial z}$$

$$\left\{ \begin{array}{l} \int \mathbf{F} \cdot d\mathbf{r} = -\Delta U \quad (\text{if } \mathbf{F} \text{ is conservative.}) \\ \int \mathbf{F} \cdot d\mathbf{r} = \Delta T \end{array} \right.$$

$$\longrightarrow \Delta T + \Delta U = 0$$

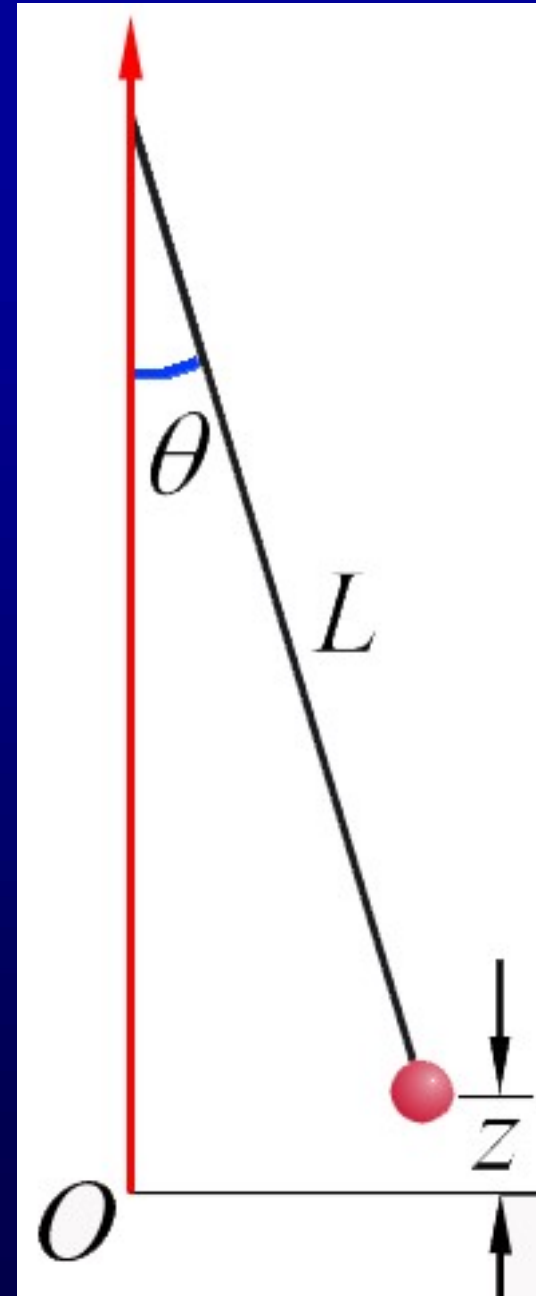
Conservation of mechanical energy

Simple pendulum

$$E = \frac{1}{2}mv^2 + mgz = mgz_0$$

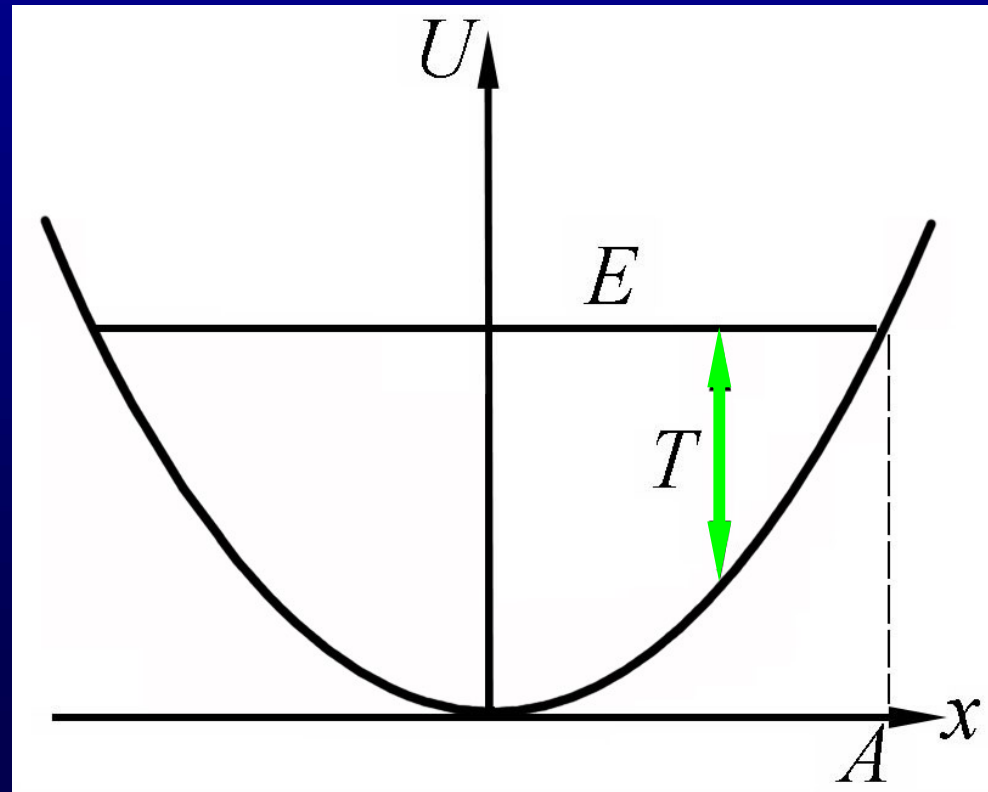
$$z = L(1 - \cos \theta)$$

$$\frac{1}{2}mv^2 = mgL(\cos \theta - \cos \theta_0)$$

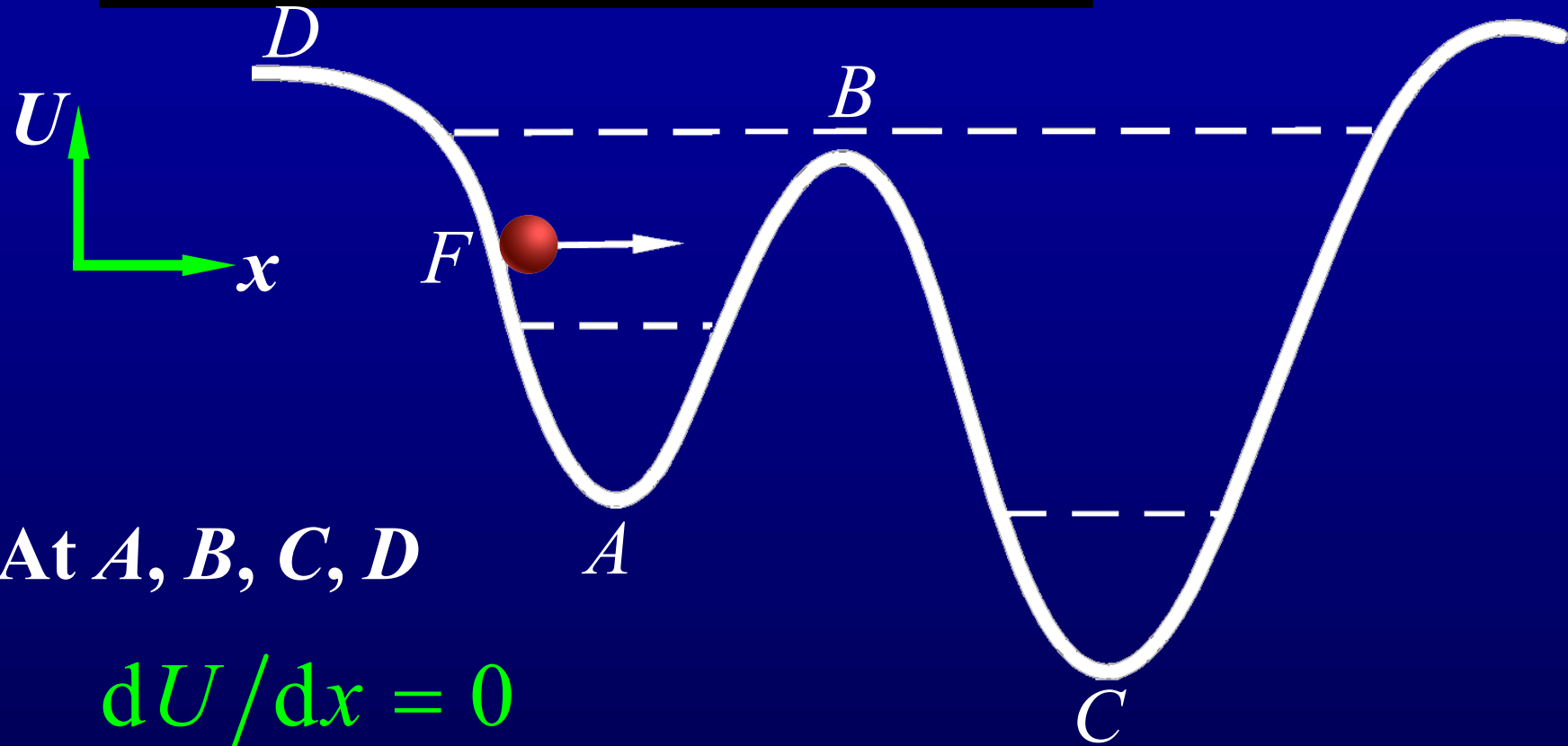


SHO(Simple harmonic oscillator)

$$\frac{1}{2}kA^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$



Model potential energy curve



At A, B, C, D

$$dU/dx = 0$$

$$F_x = 0$$

equilibrium

At F $dU/dx < 0$

$$F_x > 0$$

At A , C ,

$$dU/dx = 0$$

$$d^2U/dx^2 > 0 \longrightarrow dF/dx < 0 \text{ means ...}$$

Restoring force

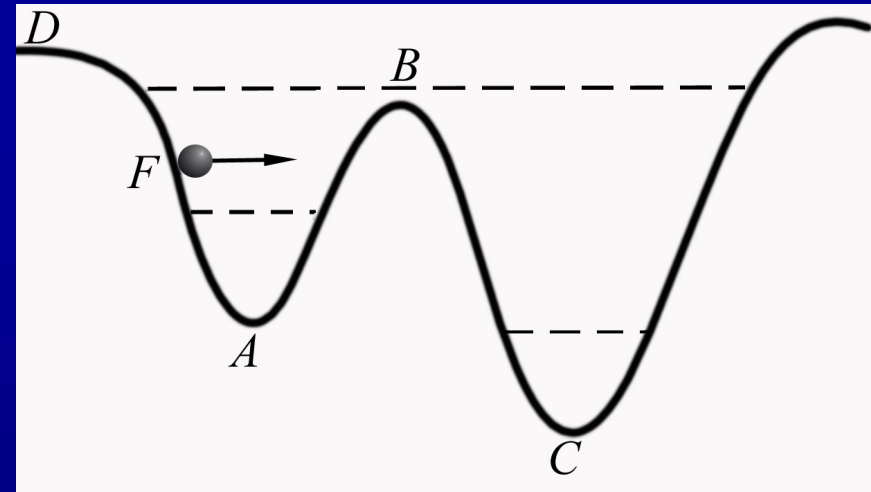
Stable equilibrium

At B ?

unstable equilibrium

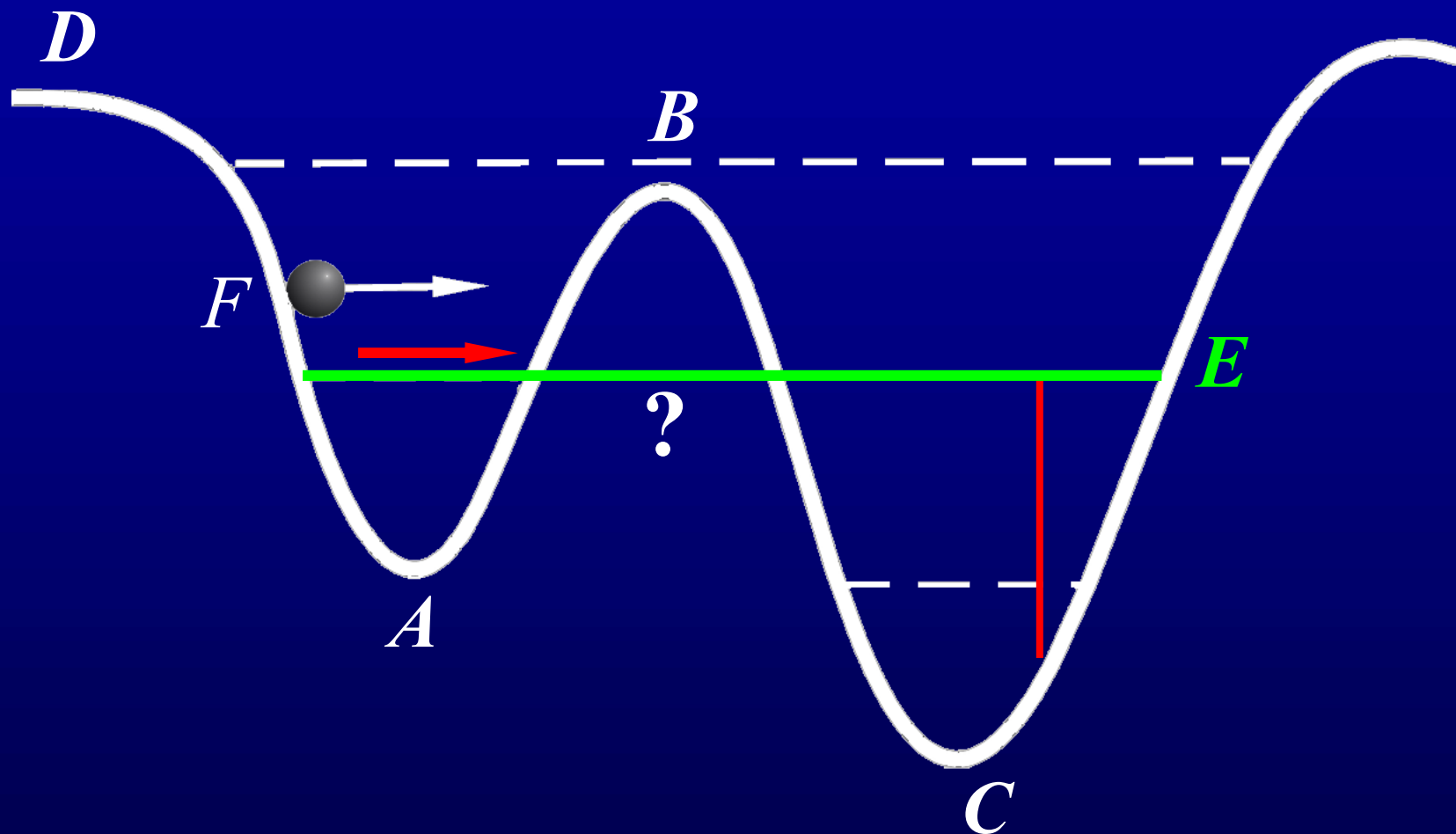
At D ?

neutral equilibrium

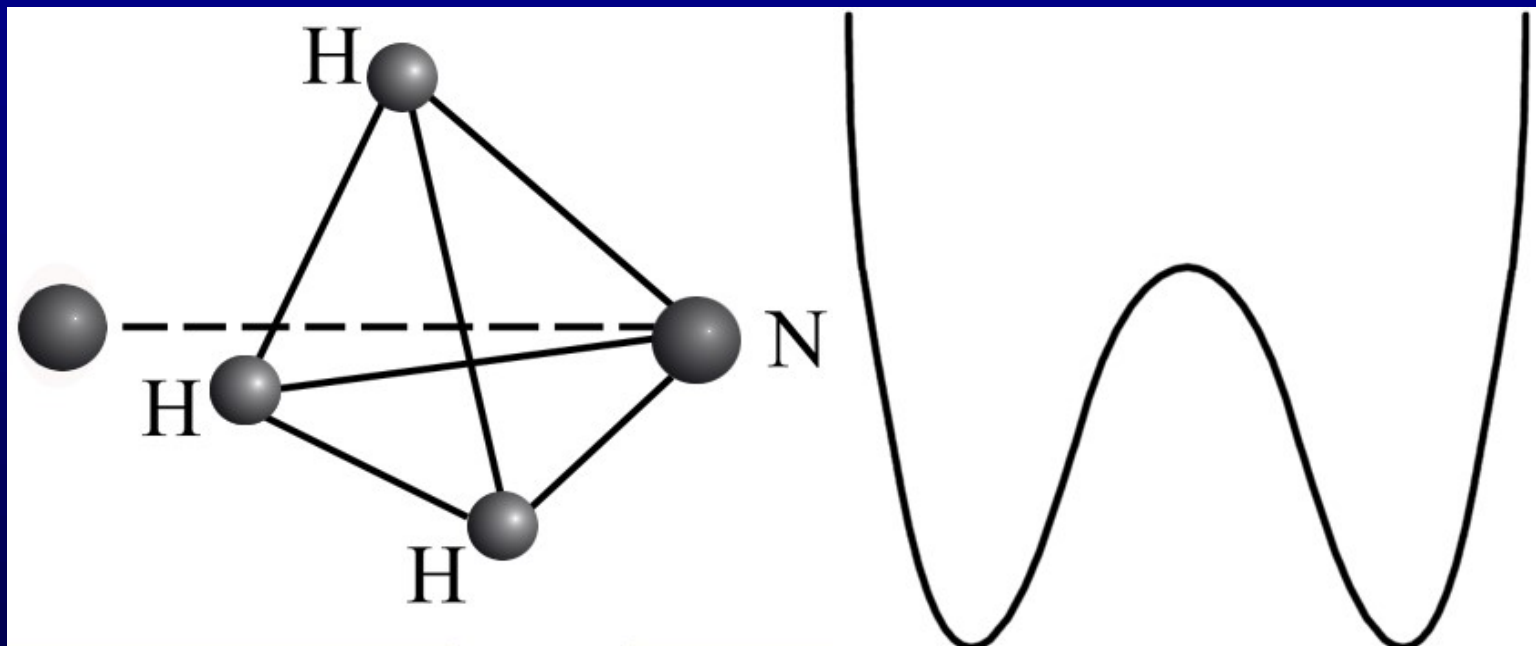


*Potential barrier and well

势垒和势阱



Ammonia molecule is an example that U



Assignment 3.10, 3.13, 3.14

Original - Photocopy of No. 0393

Abschrift/15.12.56 PM

Offener Brief an die Gruppe der Radioaktiven bei der
Gesellschafts-Tagung zu Tübingen.

Abschrift

Physikalisches Institut
der Eidg. Technischen Hochschule
Zürich

Zürich, 4. Dez. 1930
Gloriastrasse

Liebe Radioaktive Damen und Herren,

Wie der Überbringer dieser Zeilen, den ich halbvollst
anzuhören bitte, Ihnen das näherem auseinandersetzen wird, bin ich
angesichts der "falschen" Statistik der N- und Li-6 Kerne, sowie
des kontinuierlichen beta-Spektrums auf einen verzweifelten Ausweg
verfallen um den "Wechselwitz" (1) der Statistik und den Energienatz
zu retten. Nämlich die Möglichkeit, es könnten elektrisch neutrale
Teilchen, die ich Neutronen nennen will, in den Kernen existieren,
welche den Spin $1/2$ haben und das Ausschliessungsprinzip befolgen und
sich von Lichtquanten ausserdem noch dadurch unterscheiden, dass sie
nicht mit Lichtgeschwindigkeit laufen. Die Masse der Neutronen
müsste von derselben Grössenordnung wie die Elektronenmasse sein und
jedenfalls nicht grösser als 0,01 Protonenmasse.- Das kontinuierliche
beta-Spektrum wäre dann verständlich unter der Annahme, dass beim
beta-Zerfall mit dem Elektron jeweils noch ein Neutron emittiert
wird, derart, dass die Summe der Energien von Neutron und Elektron
konstant ist.

“A strange letter” by Wolfgang E. Pauli

Dear radioactive ladies and gentlemen,

As the bearer of these lines [...] will explain more exactly, considering the 'false' statistics of N-14 and Li-6 nuclei, as well as the continuous β -spectrum, I have hit upon a desperate remedy to save the "exchange theorem" of statistics and the energy theorem. Namely [there is] the possibility that there could exist in the nuclei electrically neutral particles that I wish to call neutrons,nb 2 which have spin $1/2$ and obey the exclusion principle, and additionally differ from light quanta in that they do not travel with the velocity of light: The mass of the neutron must be of the same order of magnitude as the electron mass and, in any case, not larger than 0.01 proton mass. The continuous β -spectrum would then become understandable by the assumption that in β decay a neutron is emitted together with the electron, in such a way that the sum of the energies of neutron and electron is constant.

[...]

But I don't feel secure enough to publish anything about this idea, so I first turn confidently to you, dear radioactives, with a question as to the situation concerning experimental proof of such a neutron, if it has something like about 10 times the penetrating capacity of a γ ray.

I admit that my remedy may appear to have a small *a priori* probability because neutrons, if they exist, would probably have long ago been seen. However, only those who wager can win, and the seriousness of the situation of the continuous β -spectrum can be made clear by the saying of my honored predecessor in office, Mr. Debye, [...] "One does best not to think about that at all, like the new taxes."

[...] So, dear radioactives, put it to test and set it right. [...]

With many greetings to you, also to Mr. Back, your devoted servant,

W. Pauli